

ADVANCED

Stage 6

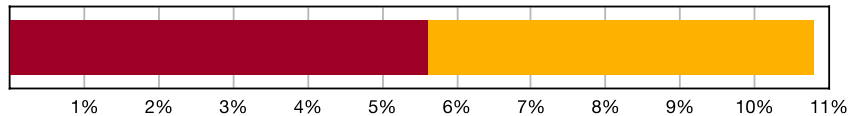
Statistics (Adv), S3 Random Variables (Adv)

Probability Density Functions (Y12)

Teacher: SHS Maths

Exam Equivalent Time: 135 minutes (based on allocation of 1.5 minutes per mark)

S3 Random Variables



- Normal Distributions
- Probability Density Functions

***SmarterMaths analytics based on the average contribution to new syllabus Advanced Maths exams since 2020.**

HISTORICAL CONTRIBUTION

- S3 Random Variables* has contributed a significant 10.8% per new syllabus Advanced exam since it was introduced in 2020.
- The topic is split into two sub-topics for the purposes of analysis: *1-Normal Distribution (5.6%)* and *2-Probability Density Functions (5.2%)*.
- This analysis looks at *Probability Density Functions*.

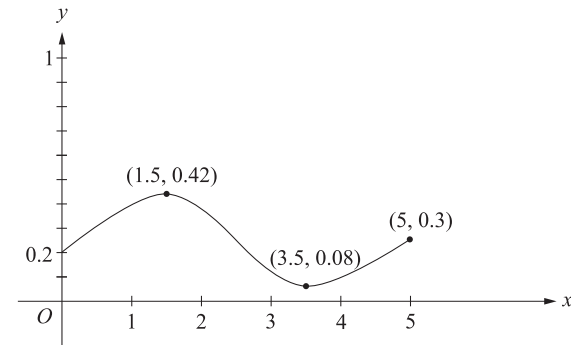
HSC ANALYSIS - What to expect and common pitfalls

- Probability Density Functions (5.2%)* has been examined in *at least* one question in each new syllabus exam to date (two in 2022 and 2021).
- The 2020-24 Advanced exams allocated 4-6 marks to this topic in each exam, producing sub-50% mean marks on every longer answer question.
- Students must be able to find the median and mode given a probability density function. Mode has been examined each year between 2021-23 and has caused major problems (see *2021 Adv 33* and *2023 Adv 29*).
- Finding mode and other probability percentiles using cumulative distributive functions is a core skill well covered in the database.
- Students must have a deep understanding of the link between a cumulative distributive functions and their probability density functions, tested three times in the last 4 years and causing major problems.

Questions

1. Statistics, 2ADV S3 EQ-Bank 6 MC

A continuous probability density function graph is drawn below.



Which of the following is the mode?

- A. 0.42
- B. 0.08
- C. 1.5
- D. 5

2. Statistics, 2ADV S3 SM-Bank 5 MC

The function $f(x)$ is a probability density function of a continuous random variable with the rule

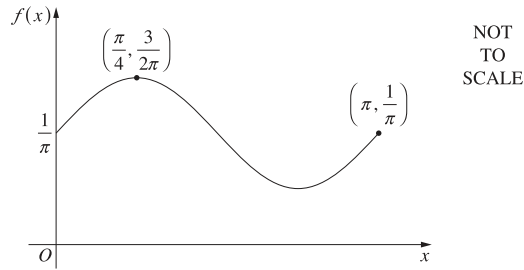
$$f(x) = \begin{cases} ae^x & 0 \leq x \leq 1 \\ ae & 1 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

The value of a is

- A. 1
- B. $\frac{1}{e}$
- C. $\frac{1}{2e}$
- D. $\frac{1}{2e-1}$

3. Statistics, 2ADV S3 2022 HSC 7 MC

Consider the following graph of a probability density function $f(x)$.



What is the value of the mode?

- A. $\frac{1}{\pi}$
- B. $\frac{3}{2\pi}$
- C. $\frac{\pi}{4}$
- D. π

4. Statistics, 2ADV S3 SM-Bank 6 MC

The continuous random variable, X , has a probability density function given by

$$f(x) = \begin{cases} \frac{1}{4} \cos\left(\frac{x}{2}\right) & 3\pi \leq x \leq 5\pi \\ 0 & \text{elsewhere} \end{cases}$$

The value of a such that $P(X < a) = \frac{\sqrt{3} + 2}{4}$ is

- A. $\frac{19\pi}{6}$
- B. $\frac{14\pi}{3}$
- C. $\frac{10\pi}{3}$
- D. $\frac{29\pi}{6}$

5. Statistics, 2ADV S3 SM-Bank 7 MC

A probability density function $f(x)$ is given by

$$f(x) = \begin{cases} \frac{1}{12}(8x - x^3) & 0 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

The median m of this function satisfies the equation

- A. $-m^4 + 16m^2 - 6 = 0$
- B. $m^4 - 16m^2 = 0$
- C. $m^4 - 16m^2 + 24 = 0.5$
- D. $m^4 - 16m^2 + 24 = 0$

6. Statistics, 2ADV S2 SM-Bank 14

A probability density function $f(x)$ is given by

$$f(x) = \begin{cases} px(3-x) & \text{if } 0 \leq x \leq 3 \\ 0 & \text{if } x < 0 \text{ or if } x > 3 \end{cases}$$

where p is a positive constant.

- i. Find the value of p . (2 marks)
- ii. Find the mode of $f(x)$. (2 marks)

7. Statistics, 2ADV S2 SM-Bank 15

A function $f(x)$ is given by

$$f(x) = \begin{cases} \frac{3}{4}(x-2)(4-x) & \text{if } 2 \leq x \leq 4 \\ 0 & \text{if } x < 2 \text{ or if } x > 4 \end{cases}$$

- i. Show this curve is a probability density function. (2 marks)
- ii. Find the mode. (2 marks)

8. Statistics, 2ADV S3 SM-Bank 20

A continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} Cx + D & 2 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

where C and D are constants.

Find the exact values of C and D , given the median of X is 4. (4 marks)

9. Statistics, 2ADV S3 SM-Bank 10

A continuous random variable, X , has a probability density function given by

$$f(x) = \begin{cases} \frac{1}{5} e^{-\frac{x}{5}} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The median of X is m .

Determine the value of m . **(3 marks)**

10. Statistics, 2ADV S3 SM-Bank 11

The probability density function of a continuous random variable X is given by

$$f(x) = \begin{cases} \frac{x}{12} & 1 \leq x \leq 5 \\ 0 & \text{otherwise} \end{cases}$$

Find $P(X < 3)$ **(2 marks)**

11. Statistics, 2ADV S3 SM-Bank 12

The function

$$f(x) = \begin{cases} k \sin(\pi x) & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

is a probability density function for the continuous random variable X .

Show that $k = \frac{\pi}{2}$. **(2 marks)**

12. Statistics, 2ADV S3 SM-Bank 2

If a continuous random variable X has probability density function

$$f(x) = \begin{cases} \frac{x}{2} & \text{if } 0 \leq x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

Find the exact value of p such that $P(X > p) = 0.4$. **(3 marks)**

13. Statistics, 2ADV S3 SM-Bank 8

The continuous random variable X has a distribution with probability density function given by

$$f(x) = \begin{cases} ax(5-x) & \text{if } 0 \leq x \leq 5 \\ 0 & \text{if } x < 0 \text{ or if } x > 5 \end{cases}$$

where a is a positive constant.

a. Find the value of a . **(3 marks)**

b. Express $P(X < 3)$ as a definite integral. (Do **not** evaluate the definite integral.) **(1 mark)**

14. Statistics, 2ADV S3 SM-Bank 9

The probability density function $f(x)$ of a random variable X is given by

$$f(x) = \begin{cases} \frac{x+1}{12} & 0 \leq x \leq 4 \\ 0 & \text{otherwise} \end{cases}$$

Find the value of b such that $P(X \leq b) = \frac{5}{8}$. **(3 marks)**

15. Statistics, 2ADV S3 2020 HSC 23

A continuous random variable, X , has the following probability density function.

$$f(x) = \begin{cases} \sin x & \text{for } 0 \leq x \leq k \\ 0 & \text{for all other values of } x \end{cases}$$

a. Find the value of k . **(2 marks)**

b. Find $P(X \leq 1)$. Give your answer correct to four decimal places. **(2 marks)**

16. Statistics, 2ADV S3 2024 HSC 25

A function $f(x)$ is defined as

$$f(x) = \begin{cases} 0, & \text{for } x < 0 \\ 1 - \frac{x}{h}, & \text{for } 0 \leq x \leq h, \\ 0, & \text{for } x > h \end{cases}$$

where h is a constant.

a. Find the value of h such that $f(x)$ is a probability density function. **(2 marks)**

b. By first finding a formula for the cumulative distribution function, sketch its graph. **(2 marks)**

c. Find the value of the median of the probability density function $f(x)$. Give your answer correct to 3 decimal places. **(2 marks)**

17. Statistics, 2ADV S3 EQ-Bank 16

A probability density function is defined by

$$f(x) = \begin{cases} a & \text{for } 0 \leq x \leq 4 \\ 3a & \text{for } 4 < x \leq 8 \end{cases}$$

- Find the value of a . **(2 marks)**
- Sketch the probability density function. **(1 mark)**
- Find an expression for the cumulative distribution function. **(2 marks)**

18. Statistics, 2ADV S3 EQ-Bank 1

A probability density function can be used to model the lifespan of a termite, X , in weeks, is given by

$$f(x) = \begin{cases} k(36 - x^2) & 3 \leq x \leq 6 \\ 0 & \text{otherwise} \end{cases}$$

- Show that the value of k is $\frac{1}{45}$. **(2 marks)**
- Find the cumulative distribution function. **(2 marks)**
- Find the probability that a termite's lifespan is greater than 5 weeks. **(1 mark)**

19. Statistics, 2ADV S3 SM-Bank 18

The Lorenz birdwing is the largest butterfly in a habitat.

The probability density function that describes its life span, X , in weeks, is given by

$$f(x) = \begin{cases} \frac{4}{625}(5x^3 - x^4) & 0 \leq x \leq 5 \\ 0 & \text{elsewhere} \end{cases}$$

In a sample of 80 Lorenz birdwing butterflies, how many butterflies are expected to live longer than two weeks, correct to the nearest integer? **(2 marks)**

20. Statistics, 2ADV S3 SM-Bank 20

A random variable X has the probability density function $f(x)$ given by

$$f(x) = \begin{cases} \frac{k}{x^2} & 1 \leq x \leq 2 \\ 0 & \text{elsewhere} \end{cases}$$

where k is a positive real number.

Show that $k = 2$. **(2 marks)**

21. Statistics, 2ADV S3 2021 HSC 30

The number of hours for which light bulbs will work before failing can be modelled by the random variable X with cumulative distribution function

$$F(x) = \begin{cases} 1 - e^{-0.01x} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

Jane sells light bulbs and promises that they will work for longer than exactly 99% of all light bulbs.

Find how long, according to Jane's promise, a light bulb bought from her should work. Give your answer in hours, rounded to two decimal places. **(2 marks)**

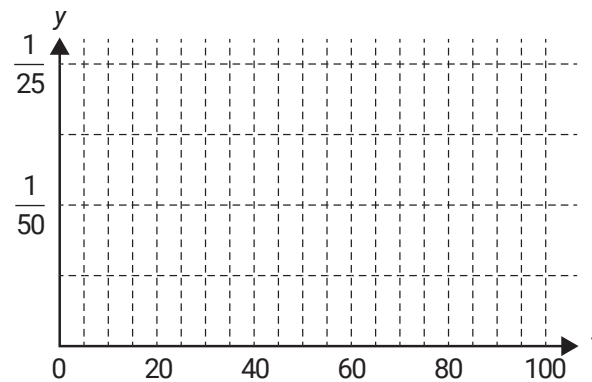
22. Statistics, 2ADV S3 SM-Bank 13

The time Jennifer spends on her homework each day varies, but she does some homework every day.

The continuous random variable T , which models the time, t , in minutes, that Jennifer spends each day on her homework, has a probability density function f , where

$$f(t) = \begin{cases} \frac{1}{625}(t - 20) & 20 \leq t < 45 \\ \frac{1}{625}(70 - t) & 45 \leq t \leq 70 \\ 0 & \text{elsewhere} \end{cases}$$

- Sketch the graph of $f(t)$ on the axes provided below. **(3 marks)**



- Find the mode. **(1 mark)**
- Find $P(25 \leq T \leq 55)$. **(2 marks)**

23. Statistics, 2ADV S3 SM-Bank 3

The continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} \pi \sin(2\pi x) & \text{if } 0 \leq x \leq \frac{1}{2} \\ 0 & \text{elsewhere} \end{cases}$$

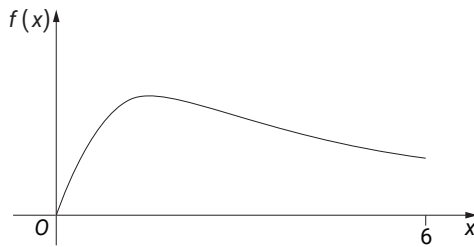
Find the value of a such that $P(X > a) = 0.2$. Give your answer to 2 decimal places. **(3 marks)**

24. Statistics, 2ADV S3 2021 HSC 33

People are given a maximum of six hours to complete a puzzle. The time spent on the puzzle, in hours, can be modelled using the continuous random variable X which has probability density function

$$f(x) = \begin{cases} \frac{Ax}{x^2 + 4} & \text{for } 0 \leq x \leq 6, (\text{ where } A > 0) \\ 0 & \text{for all other values of } x \end{cases}$$

The graph of the probability density function is shown below. The graph has a local maximum.



- Show that $A = \frac{2}{\ln 10}$. **(2 marks)**
- Show that the mode of X is two hours. **(2 marks)**
- Show that $P(X < 2) = \log_{10} 2$. **(2 marks)**
- The Intelligence Quotient (IQ) scores of people are normally distributed with a mean of 100 and standard deviation of 15.

It has been observed that the puzzle is generally completed more quickly by people with a high IQ.

It is known that 80% of people with an IQ greater than 130 can complete the puzzle in less than two hours.

A person chosen at random can complete the puzzle in less than two hours.

What is the probability that this person has an IQ greater than 130? Give your answer correct to three decimal places. **(2 marks)**

25. Statistics, 2ADV S3 2023 HSC 29

A continuous random variable X has probability density function $f(x)$ given by

$$f(x) = \begin{cases} 12x^2(1-x), & \text{for } 0 \leq x \leq 1 \\ 0, & \text{for all other values of } x \end{cases}$$

- Find the mode of X . **(2 marks)**
- Find the cumulative distribution function for the given probability density function. **(2 marks)**
- Without calculating the median, show that the mode is greater than the median. **(2 marks)**

26. Statistics, 2ADV S3 2022 HSC 30

A continuous random variable X has cumulative distribution function given by

$$F(x) = \begin{cases} 1 & x > e^3 \\ \frac{1}{k} \ln x & 1 \leq x \leq e^3 \\ 0 & x < 1 \end{cases}$$

- Show that $k = 3$. **(1 mark)**
- Given that $P(X < c) = 2P(X > c)$, find the exact value of c . **(2 marks)**

27. Statistics, 2ADV S3 SM-Bank 4

The continuous random variable X has a probability density function given by

$$f(x) = \begin{cases} \cos(2x) & \text{if } \frac{3\pi}{4} < x < \frac{5\pi}{4} \\ 0 & \text{elsewhere} \end{cases}$$

Find the value of a such that $P(X < a) = 0.25$. Give your answer correct to 2 decimal places. **(3 marks)**

Worked Solutions

1. Statistics, 2ADV S3 EQ-Bank 6 MC

Mode is the most common x -value.
 y -axis measures the probability.
Highest probability = 0.42
 $\therefore$ Mode = 1.5
 $\Rightarrow C$

2. Statistics, 2ADV S3 SM-Bank 5 MC

Total area = 1
 $\int_0^1 ae^x \, dx + \int_1^2 ae \, dx = 1$
 $[ae^x]_0^1 + [ae \cdot x]_1^2 = 1$
 $[ae - a] + [2ae - ae] = 1$
 $2ae - a = 1$
 $a(2e - 1) = 1$
 $\therefore a = \frac{1}{2e-1}$
 $\Rightarrow D$

3. Statistics, 2ADV S3 2022 HSC 7 MC

Mode $\rightarrow f(x)$ is a MAX
MAX occurs when $x = \frac{\pi}{4}$
 $\Rightarrow C$

Worked Solutions

4. Statistics, 2ADV S3 SM-Bank 6 MC

$\int_{3\pi}^a \frac{1}{4} \cos\left(\frac{x}{2}\right) dx = \frac{\sqrt{3} + 2}{4}$
 $\left[\frac{1}{2} \sin\left(\frac{x}{2}\right)\right]_{3\pi}^a = \frac{\sqrt{3} + 2}{4}$
 $\frac{1}{2} \left[\sin\left(\frac{a}{2}\right) - \sin\left(\frac{3\pi}{2}\right)\right] = \frac{\sqrt{3} + 2}{4}$
 $\frac{1}{2} \sin\left(\frac{a}{2}\right) + \frac{1}{2} = \frac{\sqrt{3} + 2}{4}$
 $\sin\left(\frac{a}{2}\right) = \frac{\sqrt{3}}{2}$
 $\frac{a}{2} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \dots$
 $\therefore a = \frac{14\pi}{3} \quad (3\pi \leq a \leq 5\pi)$
 $\Rightarrow B$

5. Statistics, 2ADV S3 SM-Bank 7 MC

$\int_0^m \frac{1}{12} (8x - x^3) dx = 0.5$
 $\left[\frac{1}{12} \left(4x^2 - \frac{x^4}{4}\right)\right]_0^m = 0.5$
 $4m^2 - \frac{m^4}{4} = 6$
 $16m^2 - m^4 = 24$
 $m^4 - 16m^2 + 24 = 0$
 $\Rightarrow D$

6. Statistics, 2ADV S2 SM-Bank 14

i. Total Area under curve = 1

$$p \int_0^3 3x - x^2 \, dx = 1$$

$$p \left[\frac{3}{2}x^2 - \frac{x^3}{3} \right]_0^3 = 1$$

$$p \left[\frac{27}{2} - \frac{27}{3} - 0 \right] = 1$$

$$\frac{9p}{2} = 1$$

$$p = \frac{2}{9}$$

ii. $f(x) = \frac{2}{9}(3x - x^2)$

$$f'(x) = \frac{2}{9}(3 - 2x)$$

S.P. when $f'(x) = 0$:

$$3 - 2x = 0$$

$$x = \frac{3}{2}$$

$$f''(x) = -\frac{4}{9} < 0 \Rightarrow \text{MAX}$$

$$\therefore \text{Mode} = \frac{3}{2}$$

7. Statistics, 2ADV S2 SM-Bank 15

i. $f(x) = \frac{3}{4}(x - 2)(4 - x)$

$$\int_2^4 f(x) = \frac{3}{4} \int_2^4 x^2 - x^2 + 6x - 8 \, dx$$

$$= \frac{3}{4} \left[\frac{-x^3}{3} + 3x^2 - 8x \right]_2^4$$

$$= \frac{3}{4} \left[\left(-\frac{64}{3} + 48 - 32 \right) - \left(-\frac{8}{3} + 12 - 16 \right) \right]$$

$$= \frac{3}{4} \left(-\frac{16}{3} + \frac{20}{3} \right)$$

$$= 1$$

$$f(2) = 0, f(4) = 0$$

$$f(x) > 0 \text{ for } 2 < x < 4$$

$$f(2) \geq 0 \text{ for } 2 \leq x \leq 4$$

$\therefore f(x)$ is a probability density function.

ii. $f(x) = \frac{3}{4}(-x^2 + 6x - 8)$

$$f'(x) = \frac{3}{4}(-2x + 6)$$

$$f''(x) = -\frac{3}{2}$$

$$\text{SP when } f'(x) = 0$$

$$-2x + 6 = 0$$

$$x = 3$$

$$f''(x) < 0 \Rightarrow \text{MAX}$$

$$\therefore \text{Mode} = 3$$

8. Statistics, 2ADV S3 SM-Bank 20

$$\int_2^5 Cx + D \, dx = 1$$
$$\left[\frac{C}{2}x^2 + Dx\right]_2^5 = 1$$
$$\left[\left(\frac{25}{2}C + 5D\right) - (2C + 2D)\right] = 1$$
$$\frac{21}{2}C + 3D = 1$$
$$21C + 6D = 2 \quad \dots \quad (1)$$

Using median $X = 4$:

$$\left[\frac{C}{2}x^2 + Dx\right]_2^4 = 0.5$$
$$[(8C + 4D) - (2C + 2D)] = 0.5$$
$$6C + 2D = 0.5 \quad \dots \quad (2)$$

Multiply: $(2) \times 3$

$$18C + 6D = 1.5 \quad \dots \quad (3)$$

Subtract: $(1) - (3)$

$$3C = \frac{1}{2}$$
$$\therefore C = \frac{1}{6}$$

Substitute into (1):

$$\frac{21}{6} + 6D = 2$$
$$6D = -\frac{9}{6}$$
$$\therefore D = -\frac{1}{4}$$

9. Statistics, 2ADV S3 SM-Bank 10

$$\frac{1}{5} \int_0^m e^{-\frac{x}{5}} dx = \frac{1}{2}$$
$$\frac{1}{5} \times (-5) \left[e^{-\frac{x}{5}}\right]_0^m = \frac{1}{2}$$
$$\left[-e^{-\frac{x}{5}}\right]_0^m = \frac{1}{2}$$
$$-e^{-\frac{m}{5}} + 1 = \frac{1}{2}$$
$$e^{-\frac{m}{5}} = \frac{1}{2}$$
$$-\frac{m}{5} = \log_e\left(\frac{1}{2}\right)$$
$$\therefore m = -5\log_e\left(\frac{1}{2}\right) \text{ (or } 5\log_e(2), \text{ or } \log_e 32)$$

10. Statistics, 2ADV S3 SM-Bank 11

$$P(X < 3)$$
$$= \int_1^3 \frac{1}{12} x dx$$
$$= \frac{1}{12} \left[\frac{1}{2}x^2\right]_1^3$$
$$= \frac{1}{24} [3^2 - 1^2]$$
$$= \frac{1}{3}$$

11. Statistics, 2ADV S3 SM-Bank 12

Total Area under curve = 1 u²

$$\int_0^1 k \sin(\pi x) \, dx = 1$$
$$-\frac{k}{\pi} [\cos(\pi x)]_0^1 = 1$$
$$-\frac{k}{\pi} [\cos(\pi) - \cos(0)] = 1$$
$$-\frac{k}{\pi} [-1 - 1] = 1$$
$$2k = \pi$$
$$\therefore k = \frac{\pi}{2}$$

12. Statistics, 2ADV S3 SM-Bank 2

$$\int_p^2 \frac{x}{2} dx = 0.4$$

$$\left[\frac{x^2}{4} \right]_p^2 = \frac{2}{5}$$

$$1 - \frac{p^2}{4} = \frac{2}{5}$$

$$\frac{p^2}{4} = \frac{3}{5}$$

$$p^2 = \frac{12}{5}$$

$$\therefore p = \frac{2\sqrt{3}}{\sqrt{5}}$$

$$= \frac{2\sqrt{15}}{5} \quad (0 \leq p \leq 2)$$

13. Statistics, 2ADV S3 SM-Bank 8

a. Total Area under curve = 1

$$a \int_0^5 (5x - x^2) dx = 1$$

$$a \left[\frac{5}{2}x^2 - \frac{1}{3}x^3 \right]_0^5 = 1$$

$$a \left[\left(\frac{125}{2} - \frac{125}{3} \right) - (0) \right] = 1$$

$$\frac{125}{6}a = 1$$

$$\therefore a = \frac{6}{125}$$

b. $P(X < 3) = \int_0^3 ax(5-x) dx$

14. Statistics, 2ADV S3 SM-Bank 9

$$\frac{1}{12} \int_0^b (x+1) dx = \frac{5}{8}$$

$$\left[\frac{1}{2}x^2 + x \right]_0^b = \frac{15}{2}$$

$$\frac{1}{2}b^2 + b = \frac{15}{2}$$

$$b^2 + 2b - 15 = 0$$

$$(b+5)(b-3) = 0$$

$$\therefore b = 3 \quad (0 \leq b \leq 4)$$

15. Statistics, 2ADV S3 2020 HSC 23

a. $\int_0^k \sin x = 1$

$$[-\cos x]_0^k = 1$$

$$-\cos k + \cos 0 = 1$$

$$-\cos k = 0$$

$$\cos k = 0$$

$$k = \frac{\pi}{2}$$

b. $P(X \leq 1) = \int_0^1 \sin x dx$

$$= [-\cos x]_0^1$$

$$= -\cos 1 + \cos 0$$

$$= 1 - \cos 1$$

$$= 0.45969\dots$$

$$= 0.4597 \quad (\text{to 4 d.p.})$$

♦ Mean mark part (b) 44%.

16. Statistics, 2ADV S3 2024 HSC 25

a. $f(x)$ is a PDF if:

$$\int_0^h 1 - \frac{x}{h} dx = 1$$

$$\left[x - \frac{x^2}{2h} \right]_0^h = 1$$

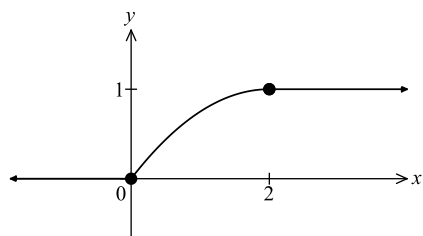
$$h - \frac{h}{2} = 1$$

$$h = 2$$

b. $\int 1 - \frac{x}{2} dx = x - \frac{x^2}{4} + c$

At $x = 0$, $F(0) = 0$, $c = 0$

$$f(x) = \begin{cases} 0, & \text{for } x < 0 \\ x - \frac{x^2}{4}, & \text{for } 0 \leq x \leq 2, \\ 1, & \text{for } x > 2 \end{cases}$$



c. Let $m = \text{median of } f(x)$

$$\int_0^m 1 - \frac{x}{2} dx = 0.5$$

$$\left[x - \frac{x^2}{4} \right]_0^m = 0.5$$

$$m - \frac{m^2}{4} = 0.5$$

$$4m - m^2 = 2$$

$$m^2 - 4m + 2 = 0$$

$$(m - 2)^2 = 2$$

$$m = 2 \pm \sqrt{2}$$

$$\therefore \text{Median} = 2 - \sqrt{2} \quad (x \in [0, 2])$$

$$= 0.586 \text{ (3 d.p.)}$$

♦♦ Mean mark (b) 27%.

♦♦ Mean mark (c) 38%.

17. Statistics, 2ADV S3 EQ-Bank 16

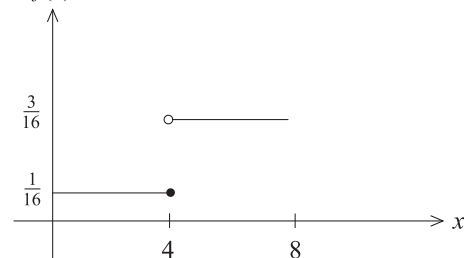
i. $\int_0^4 a dx = [ax]_0^4 = 4a$

$$\int_4^8 3a dx = [3ax]_4^8 = 24a - 12a = 12a$$

$$4a + 12a = 1$$

$$a = \frac{1}{16}$$

ii. $f(x)$



iii. When $0 \leq x \leq 4$:

$$F(x) = \int \frac{1}{16} dx = \frac{x}{16} + C$$

$$F(0) = 0 \Rightarrow C = 0$$

$$F(x) = \frac{x}{16} \dots (1)$$

When $4 < x \leq 8$:

$$F(x) = \int \frac{3}{16} dx = \frac{3x}{16} + C$$

$$F(4) = \frac{1}{4} \text{ (see (1) above)}$$

$$\frac{3 \times 4}{16} + C = \frac{1}{4}$$

$$C = -\frac{1}{2}$$

$$F(x) = \frac{3x}{16} - \frac{1}{2}$$

$$\therefore F(x) = \begin{cases} \frac{x}{16} & \text{for } 0 \leq x \leq 4 \\ \frac{3x}{16} - \frac{1}{2} & \text{for } 4 < x \leq 8 \end{cases}$$

COMMENT: Understand why you can check your equation here by confirming $F(8) = 1$

18. Statistics, 2ADV S3 EQ-Bank 1

$$\begin{aligned} \text{i.} \quad k \int_3^6 36-x^2 \, dx &= 1 \\ k \left[36x - \frac{x^3}{3} \right]_3^6 &= 1 \\ k[(216-72) - (108-9)] &= 1 \\ 45k &= 1 \\ k &= \frac{1}{45} \end{aligned}$$

$$\begin{aligned} \text{ii.} \quad F(t) &= \int_{-\infty}^t f(x) \, dx \\ &= \int_3^t f(x) \, dx \\ &= \frac{1}{45} \int_3^t 36-x^2 \, dx \\ &= \frac{1}{45} \left[36x - \frac{x^3}{3} \right]_3^t \\ &= \frac{1}{135} [108x - x^3]_3^t \\ &= \frac{1}{35} [(108t - t^3) - (324 - 27)] \\ &= \frac{1}{135} (108t - t^3 - 297) \end{aligned}$$

$$\therefore F(x) = \begin{cases} 0 & x < 3 \\ \frac{1}{135} (108t - t^3 - 297) & 3 \leq x \leq 6 \\ 1 & x > 6 \end{cases}$$

$$\begin{aligned} \text{iii.} \quad P(X > 5) &= 1 - F(5) \\ &= 1 - \frac{1}{135} (108 \times 5 - 5^3 - 297) \\ &= 1 - \frac{118}{135} \\ &= \frac{17}{135} \end{aligned}$$

19. Statistics, 2ADV S3 SM-Bank 18

$$\begin{aligned} \Pr(X > 2) &= \frac{4}{625} \int_2^5 5x^3 - x^4 \, dx \\ &= \frac{4}{625} \left[\frac{5}{4} x^4 - \frac{x^5}{5} \right]_2^5 \\ &= \frac{4}{625} \left[\left(\frac{5^5}{4} - \frac{5^5}{5} \right) - \left(\frac{5}{4} \times 2^4 - \frac{2^5}{5} \right) \right] \\ &= \frac{4}{625} \left[\frac{625}{4} - \frac{68}{5} \right] \\ &= 0.9129 \dots \end{aligned}$$

$$\begin{aligned} \therefore \text{Expected number} &= 80 \times 0.9129 \dots \\ &\approx 73.03 \\ &\approx 73 \end{aligned}$$

20. Statistics, 2ADV S3 SM-Bank 20

$$\begin{aligned} \int_1^2 \frac{k}{x^2} \, dx &= 1 \\ k \left[-\frac{1}{x} \right]_1^2 &= 1 \\ k \left(-\frac{1}{2} + 1 \right) &= 1 \\ \frac{k}{2} &= 1 \\ \therefore k &= 2 \quad \dots \text{ as required} \end{aligned}$$

◆ Mean mark 48%.

21. Statistics, 2ADV S3 2021 HSC 30

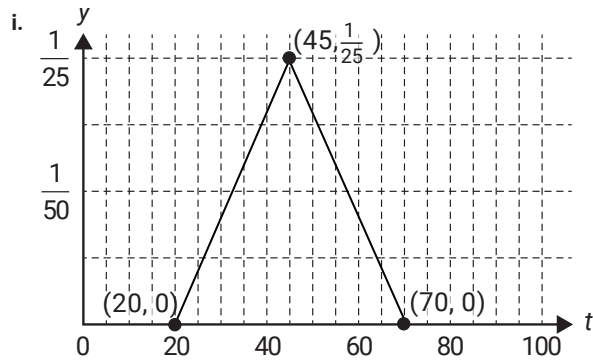
Find x when $F(x) > 0.99$:

$$\begin{aligned} 1 - e^{-0.01x} &= 0.99 \\ e^{-0.01x} &= 0.01 \\ -0.01x &= \ln(0.01) \\ x &= \frac{\ln(0.01)}{-0.01} \\ &= 460.517 \dots \\ &= 460.52 \text{ hours} \end{aligned}$$

$\therefore$ Light bulbs should work at least 460.52 hours.

◆◆ Mean mark 22%.

22. Statistics, 2ADV S3 SM-Bank 13



ii. Mode = 45 (value of t at highest value of $f(t)$)

iii. $P(25 \leq T \leq 55)$

$$\begin{aligned}
 &= \int_{25}^{45} \frac{1}{625} (t - 20) dt + \int_{45}^{55} \frac{1}{625} (70 - t) dt \\
 &= \frac{1}{625} \left[\frac{t^2}{2} - 20t \right]_{25}^{45} + \frac{1}{625} \left[70t - \frac{t^2}{2} \right]_{45}^{55} \\
 &= \frac{1}{625} [112.5 - (-187.5)] + \frac{1}{625} (2337.5 - 2137.5) \\
 &= \frac{300}{625} + \frac{200}{625} \\
 &= \frac{4}{5}
 \end{aligned}$$

MARKER'S COMMENT: Many did not draw graph along t -axis between 0 and 20 and for $t > 70$.

23. Statistics, 2ADV S3 SM-Bank 3

$$\int_a^{\frac{1}{2}} \pi \sin(2\pi x) dx = 0.2$$

$$-\frac{1}{2} [\cos(2\pi x)]_a^{\frac{1}{2}} = 0.2$$

$$-\frac{1}{2} [\cos(\pi) - \cos(2\pi a)] = 0.2$$

$$-\frac{1}{2} (-1 - \cos(2\pi a)) = 0.2$$

$$-1 - \cos(2\pi a) = -0.4$$

$$\cos(2\pi a) = -0.6$$

$$2\pi a = \cos^{-1}(-0.6)$$

$$\therefore a = \frac{\cos^{-1}(-0.6)}{2\pi}$$

$$= 0.3524\dots$$

$$= 0.35 \quad (\text{to 2 d.p.})$$

24. Statistics, 2ADV S3 2021 HSC 33

a. $\int_0^6 \frac{Ax}{x^2 + 4} dx = 1$

$$\frac{A}{2} \int_0^6 \frac{2x}{x^2 + 4} dx = 1$$
$$\frac{A}{2} [\ln(x^2 + 4)]_0^6 = 1$$
$$\frac{A}{2} (\ln 40 - \ln 4) = 1$$
$$\frac{A}{2} \ln\left(\frac{40}{4}\right) = 1$$
$$\frac{A}{2} \ln 10 = 1$$
$$A = \frac{2}{\ln 10}$$

b. Mode $\rightarrow f(x)$ is a MAX

$$f(x) = \frac{Ax}{x^2 + 4}$$
$$f'(x) = \frac{A(x^2 + 4) - Ax(2x)}{(x^2 + 4)^2}$$
$$= \frac{Ax^2 + 4A - 2Ax^2}{(x^2 + 4)^2}$$
$$= \frac{A(4 - x^2)}{(x^2 + 4)^2}$$

$f(x)$ max occurs when $f'(x) = 0$:

$$4 - x^2 = 0$$
$$x = 2 \quad (x > 0)$$

c. $P(X < 2) = \int_0^2 \frac{Ax}{x^2 + 4} dx$

$$= \frac{A}{2} [\ln(x^2 + 4)]_0^2$$
$$= \frac{1}{\ln 10} (\ln 8 - \ln 4)$$
$$= \frac{1}{\ln 10} \left(\ln \frac{8}{4}\right)$$
$$= \frac{1}{\ln 10} \cdot \ln 2$$
$$= \log_{10} 2$$

d. $z\text{-score}(130) = \frac{x - \mu}{\sigma} = \frac{130 - 100}{15} = 2$

$$P(z > 2) = 2.5\%$$

◆◆◆ Mean mark part (b) 24%.

◆◆ Mean mark part (c) 30%.

◆◆ Mean mark part (d) 25%.

$$P(\text{IQ} > 130 \mid x < 2) = \frac{P(\text{IQ} > 130 \cap X < 2)}{P(X < 2)}$$
$$= \frac{0.8 \times 0.025}{\log_{10} 2}$$
$$= 0.0664 \dots$$
$$= 0.066$$

25. Statistics, 2ADV S3 2023 HSC 29

a. Mode $\rightarrow f(x)_{\text{max}}$

$$f(x) = 12x^2(1 - x) = 12x^2 - 12x^3$$
$$f'(x) = 24x - 36x^2 = 12x(2 - 3x)$$
$$f''(x) = 24 - 72x$$

Max/min when $f'(x) = 0$

$$2 - 3x = 0 \Rightarrow x = \frac{2}{3} \quad (x \neq 0)$$

$$\text{At } x = \frac{2}{3}, f''(x) = 24 - 72\left(\frac{2}{3}\right) = -24 < 0$$

$$\therefore \text{Mode (max) at } x = \frac{2}{3}$$

b. $F(x) = \int 12x^2 - 12x^3 dx$

$$= 4x^3 - 3x^4 + c$$

$$\text{At } x = 0, F(x) = 0 \Rightarrow c = 0$$

$$F(x) = 4x^3 - 3x^4$$

$$F(x) = \begin{cases} 0, & \text{for } x < 0 \\ 4x^3 - 3x^4, & \text{for } 0 \leq x \leq 1 \\ 1, & \text{for } x > 1 \end{cases}$$

c. Find $F\left(\frac{2}{3}\right)$:

$$F\left(\frac{2}{3}\right) = 4 \times \left(\frac{2}{3}\right)^3 - 3 \times \left(\frac{2}{3}\right)^4$$
$$= \frac{16}{27} > 0.5$$

$$\therefore \text{Mode} > \text{median}$$

◆ Mean mark (a) 45%.

◆◆◆ Mean mark (c) 17%.

26. Statistics, 2ADV S3 2022 HSC 30

a. Show $k = 3$

$$\begin{aligned} \left[\frac{1}{k}\ln x\right]_1^{e^3} &= 1 \\ \frac{1}{k}\ln(e^3) - \frac{1}{k}\ln 1 &= 1 \\ \frac{1}{k}(3) - \frac{1}{k}(0) &= 1 \\ k &= 3 \dots \text{as required} \end{aligned}$$

♦♦ Mean mark (a) 38%.

b. $P(X < c) = \left[\frac{1}{3}\ln x\right]_0^c$

$$= \frac{1}{3}\ln c$$

$$\begin{aligned} 2P(X > c) &= 2P(1 - P(X < c)) \\ &= 2\left(1 - \frac{1}{3}\ln c\right) \end{aligned}$$

Given $P(X < c) = 2P(X > c)$

$$\begin{aligned} \frac{1}{3}\ln c &= 2 - \frac{2}{3}\ln c \\ \ln c &= 2 \\ \therefore c &= e^2 \end{aligned}$$

♦♦♦ Mean mark (b) 19%.

27. Statistics, 2ADV S3 SM-Bank 4

$$\begin{aligned} \int_{\frac{3\pi}{4}}^a \cos(2x)dx &= 0.25 \\ \frac{1}{2}[\sin(2x)]_{\frac{3\pi}{4}}^a &= 0.25 \\ \left[\sin(2a) - \sin\left(\frac{3\pi}{2}\right)\right] &= 0.5 \\ \sin(2a) + 1 &= 0.5 \\ 2a &= \sin^{-1}(-0.5) \\ &= -0.5235 \end{aligned}$$

Since sin is negative in 3rd/4th quadrants:

$$\begin{aligned} 2a &= \pi + 0.5234 \\ a &= 1.832\dots \quad (\text{not in range}) \\ \text{or} \\ 2a &= 2\pi - 0.5234 \\ a &= 2.87989\dots \\ &= 2.88 \quad \left(\text{in range: } \frac{3\pi}{4} < x < \frac{5\pi}{4}\right) \end{aligned}$$